Important information. Please read carefully.

ZYLOVAA™-H

LOSARTAN POTASSIUM 50MG / HYDROCHLOROTHIAZIDE 12.5MG
LOSARTAN POTASSIUM 100MG / HYDROCHLOROTHIAZIDE 25MG
Film-Coated Tablet

COMPOSITION

Zylovaa-H Tablet 50/12.5mg:
Each film-coated tablet contains 50mg Losartan Potassium and 12.5mg Hydrochlorothiazide
Zylovaa -H Tablet 100/25mg:
Each film-coated tablet contains 100mg Losartan Potassium and 25mg Hydrochlorothiazide

PHARMACODYNAMICS

The components of losartan-hydrochlorothiazide have been shown to have an additive effect on blood-pressure reduction, reducing blood pressure to a greater degree than either component alone. This effect is thought to be a result of the complimentary actions of both components. Further, as a result of its diuretic effect, hydrochlorothiazide increases plasma-renin activity, increases aldosterone secretion, decreases serum potassium, and increases the levels of angiotensin II. Administration of losartan blocks all the physiologically relevant actions of angiotensin II and through inhibition of aldosterone could tend to attenuate the potassium loss associated with the diuretic.
Losartan has been shown to have a mild and transient uricosuric effect. Hydrochlorothiazide has been shown to cause modest increases in uric acid; the combination of losartan and hydrochlorothiazide tends to attenuate the diuretic-induced hyperuricaemia.
Non-melanoma skin cancer: Based on available data from epidemiological studies, cumulative dose-dependent association between hydrochlorothiazide and NMSC has been observed. One study included a population comprised of 71,533 cases of BCC and of 8,629 cases of SCC matched to 1,430,833 and 172,462 population controls, respectively. High hydrochlorothiazide use (>50,000 mg cumulative) was associated with an adjusted OR of 1.29 (95% CI: 1.23-1.35) for BCC and 3.98 (95% CI: 3.68-4.31) for SCC. A clear cumulative dose response relationship was observed for both BCC and SCC. Another study showed a possible association between lip cancer (SCC) and exposure to hydrochlorothiazide: 633 cases of lip-cancer were matched with 63,067 population controls, using a risk-set sampling strategy. A cumulative dose-response relationship was demonstrated with an adjusted OR 2.1 (95% CI: 1.7-2.6) increasing to OR 3.9 (3.0-4.9) for high use (~25,000 mg) and OR 7.7 (5.7-10.5) for the highest cumulative dose (~100,000 mg).

PHARMACOKINETICS

Absorption
Losartan
Following oral administration, losartan is well absorbed and undergoes first-pass metabolism, forming an active carboxylic acid metabolite and other inactive metabolites. The systemic bioavailability of losartan tablets is approximately 33%. Mean peak concentrations of losartan and its active metabolite are reached in 1 hour and in 3-4 hours, respectively. There was no clinically significant effect on the plasma concentration profile of losartan when the drug was administered with a standardised meal.

Distribution
Losartan
Both losartan and its active metabolite are ≥ 99% bound to plasma proteins, primarily albumin. The volume of distribution of losartan is 34 litres.

Hydrochlorothiazide
Hydrochlorothiazide crosses the placental but not the blood-brain barrier and is excreted in breast milk.

Metabolism
Losartan
About 14% of an intravenously or orally-administered dose of losartan is converted to its active metabolite. In addition to the active metabolite, inactive metabolites are formed, including two major metabolites formed by hydroxylation of the butyl side chain and a minor metabolite, an N-2 tetrazole glucuronide.

Elimination
Losartan
Plasma clearance of losartan and its active metabolite is about 600 ml/min and 50 ml/min, respectively. Renal clearance of losartan and its active metabolite is about 74 ml/min and 26 ml/min, respectively. When losartan is administered orally, about 4% of the dose is excreted unchanged in the urine, and about 6% of the dose is excreted in the urine as active metabolite. The pharmacokinetics of losartan and its active metabolite are linear with oral losartan potassium doses up to 200mg.
Following oral administration, plasma concentrations of losartan and its active metabolite decline polyexponentially with a terminal half-life of about 2 hours and 6-9 hours, respectively. During once-daily dosing with 100 mg, neither losartan nor its active metabolite accumulates significantly in plasma.
Both biliary and urinary excretions contribute to the elimination of losartan and its metabolites.
Hydrochlorothiazide
Hydrochlorothiazide is not metabolised but is eliminated rapidly by the kidney. When plasma levels have been followed for at least 24 hours, the plasma half-life has been observed to vary between 5.6 and 14.8 hours. At least 61% of the oral dose is eliminated unchanged within 24 hours.

Losartan-Hydrochlorothiazide
The plasma concentrations of losartan and its active metabolite and the absorption of hydrochlorothiazide in elderly hypertensives are not significantly different from those in young hypertensives.
Losartan
Following oral administration in patients with mild to moderate alcoholic cirrhosis of the liver, plasma concentrations of losartan and its active metabolite were, respectively, 5-fold and 1.7-fold greater than those seen in young male volunteers. Plasma concentrations of losartan are not altered in patients with creatinine clearance above 10 ml/min.
Neither losartan nor the active metabolite can be removed by haemodialysis.

INDICATIONS

Zylovaa-H is indicated for the treatment of hypertension, for patients in whom combination therapy is appropriate.
In hypertensive patients with left ventricular hypertrophy, a reduced risk of stroke was demonstrated with losartan administered usually in combination with hydrochlorothiazide.

DOSAGE AND ADMINISTRATION

To be administered orally.
- *Zylovaa-H* may be administered with or without food.
- *Zylovaa-H* may be administered with other antihypertensive agents.
- Fixed dose combination is not indicated for initial therapy.

Hypertension
The usual starting and maintenance dose is 1 tablet Zylovaa-H 50/12.5mg (losartan potassium 50mg/ hydrochlorothiazide 12.5mg) once daily for most patients. For patients who do not respond adequately to Zylovaa-H 50/12.5mg, the dosage may be increased to 1 tablet Zylovaa-H 100/25mg (losartan potassium 100mg/ hydrochlorothiazide 25mg) once daily or 2 tablets Zylovaa-H 50/12.5mg once daily. The maximum dose is 1 tablet Zylovaa-H 100/25mg once daily or 2 tablets Zylovaa-H 50/12.5mg once daily. In general, the antihypertensive effect is attained within three weeks after initiation of therapy.
Zylovaa-H should not be initiated in patients who are intravascularly volume-depleted (e.g. those treated with high-dose diuretics).
Zylovaa-H is not recommended for patients with severe renal impairment (creatinine clearance ≤ 30ml/min) or for patients with hepatic impairment.
No initial dosage adjustment of Zylovaa-H 50/12.5mg is necessary for elderly patients. Zylovaa-H Tablet 100/25mg should not be used as initial therapy in elderly patients.

Hypertensive patients with left ventricular hypertrophy
The usual starting dose is 50mg of losartan once daily. If goal blood pressure is not reached with losartan 50mg, therapy should be titrated using a combination of losartan and a low dose of hydrochlorothiazide (12.5mg) and, if needed the dose should then be increased to losartan 100mg and hydrochlorothiazide 12.5mg once daily. If necessary, the dose should be increased to losartan 100mg and hydrochlorothiazide 25mg once daily. Zylovaa-H 50/12.5mg and Zylovaa-H 100/25mg are suitable alternative formulations in patients who would otherwise be treated concomitantly with losartan plus hydrochlorothiazide.

CONTRAINDICATIONS

Zylovaa-H is contraindicated in:
• Patients who are hypersensitive to any component of this product.
• Patients with anuria.
• Patients who are hypersensitive to other sulfonamide-derived drugs.
• The concomitant use of ZYLOVAA-H TABLET (i.e composed of Losartan Potassium & Hydrochlorothiazide) with aliskiren-containing products is contraindicated in patients with diabetes mellitus or renal impairment (GFR < 60 ml/min/1.73 m^2).

WARNINGS AND PRECAUTIONS

Losartan-hydrochlorothiazide
Hypersensitivity: Angioedema (See Side Effects)
Hepatic and renal impairment: Losartan-hydrochlorothiazide is not recommended for patients with hepatic impairment or severe renal impairment (creatinine clearance ≤ 30ml/min). (See Dosage and Administration)

Losartan
Renal function Impairment
As a consequence of inhibiting the renin-angiotensin-aldosterone system, changes in renal function including renal failure have been reported in susceptible individuals; as with other drugs that affect the renin-angiotensin-aldosterone system, increases in blood urea and serum creatinine have also been reported in patients with bilateral renal artery stenosis or stenosis of the artery to a solitary kidney. These changes in renal function may be reversible upon discontinuation of therapy.

Hydrochlorothiazide
Hypotension and electrolyte/fluid imbalance: Symptomatic hypotension may occur in some patients. Patients should be observed for clinical signs of fluid or electrolyte imbalance, e.g. volume depletion, hyponatremia, hypochloremic alkalosis, hypomagnesemia or hypokalemia which may occur during intercurrent diarrhea or vomiting. Periodic determination of serum electrolytes should be performed at appropriate intervals in such patients.
Metabolic and endocrine effects: Thiazide therapy may impair glucose tolerance. Dosage adjustment of antidiabetic agents, including Insulin, may be required. (See Drug Interactions).
Thiazides may decrease urinary calcium excretion and may cause intermittent and slight elevation of serum calcium. Marked hypercalcaemia may be evidence of hidden hyperparathyroidism. Thiazides should be discontinued before carrying out tests for parathyroid function.
Increases in cholesterol and triglyceride levels may be associated with thiazide diuretic therapy. Thiazide therapy may precipitate hyperuricaemia and/or gout in certain patients. Because losartan decreases uric acid, losartan in combination with hydrochlorothiazide attenuates the diuretic-induced hyperuricaemia.
Other: In patients receiving thiazides, hypersensitivity reactions may occur with or without a history of allergy or bronchial asthma. Exacerbation or activation of systemic lupus erythematosus has been reported with the use of thiazides.
An increased risk of non-melanoma skin cancer (NMSC) [basal cell carcinoma (BCC) and squamous cell carcinoma (SCC)] with increasing cumulative dose of hydrochlorothiazide exposure has been observed in two epidemiological studies based on the Danish National Cancer Registry. Photosensitizing actions of hydrochlorothiazide could act as a possible mechanism for NMSC.
Patients taking hydrochlorothiazide should be informed of the risk of NMSC and advised to regularly check their skin for any new lesions and promptly report any suspicious skin lesions. Possible preventive measures such as limited exposure to sunlight and UV rays and, in case of exposure, adequate protection should be advised to the patients in order to minimize the risk of skin cancer. Suspicious skin lesions should be promptly examined potentially including histological examinations of biopsies. The use of hydrochlorothiazide may also need to be reconsidered in patients who have experienced previous NMSC.

DRUG INTERACTIONS

Losartan
No drug Interactions have been identified with hydrochlorothiazide, digoxin, warfarin, cimetidine, ketoconazole, erythromycin and phenobarbital (phenobarbitone). Rifampicin and fluconazole have been reported to reduce levels of active metabolite. The clinical consequences of these interactions have not been evaluated.
As with other drugs that block angiotensin II or its effects, concomitant use of potassium-sparing diuretics (e.g. spironolactone, triamterene, amiloride), potassium supplements, or salt substitutes containing potassium may lead to increases in serum potassium.
As with other drugs which affect the excretion of sodium, lithium excretion may be reduced. Therefore, serum lithium levels should be monitored carefully if lithium salts are to be co-administered with angiotensin II receptor antagonists.
As with other antihypertensive agents, the antihypertensive effect of losartan may be attenuated by the non-steroidal anti-inflammatory drugs (NSAIDs) including selective cyclooxygenase-2 inhibitors (COX-2 Inhibitors).
In some patients with compromised renal function who are being treated with non-steroidal anti-inflammatory drugs (NSAIDs) including selective cyclooxygenase-2 inhibitors (COX-2 inhibitors), co-administered with angiotensin II receptor antagonists may result in a further deterioration of renal function. These effects are usually reversible.

Hydrochlorothiazide
When given concurrently, the following drugs may interact with thiazide diuretics:
Alcohol barbiturates, or narcotics- potentiation of orthostatic hypotension may occur.
Antidiabetic drugs (oral agents and insulin)- dosage adjustment of the antidiabetic drug may be required.
Other antihypertensive drugs-there may be an additive effect.
Colestyramine and colestipol resins- absorption of hydrochlorothiazide is impaired in the presence of anionic exchange resins. Single doses of either colestyramine or colestipol resins bind the hydrochlorothiazide and reduce its absorption from the gastro-intestinal tract by up to 85% and 43%, respectively.
Corticosteroids, ACTH- intensified electrolyte depletion, particularly hypokalaemia.
Pressor amines (e.g. adrenaline)- possible decreased response to pressor amines, but not sufficient to preclude their use.
Skeletal muscle relaxants, non-depolarising (e.g. tubocurarine)- possible increased responsiveness to the muscle relaxant.
Lithium- diuretic agents reduce the renal clearance of lithium and add a high risk of lithium toxicity; concomitant use is not recommended. Refer to the prescribing information for lithium preparations before use of such preparations.
Non-steroidal anti-inflammatory drugs including selective cyclooxygenose-2 inhibitors (COX-2 inhibitors)- in some patients, the administration of a non-steroidal anti-inflammatory agent including selective COX-2 inhibitors can reduce the diuretic, natriuretic, and antihypertensive effects of diuretics.
Drug/laboratory test *Interactions*
Because of their effects on calcium metabolism, thiazides may interfere with tests for parathyroid function. (See Warnings and Precautions).

PREGNANCY AND LACTATION

Use during pregnancy
When used in pregnancy during the second and third trimesters, drugs that act directly on the renin-angiotensin system can cause injury and even death to the developing foetus. When pregnancy is detected, losartan-hydrochlorothiazide should be discontinued as soon as possible.
Use during lactation
It is not known whether losartan is excreted in human milk. Thiazides appear in human milk. Because of the potential for adverse effects on the breast-feeding infant, a decision should be made whether to discontinue breast-feeding or discontinue the drug, taking into account the importance of the drug to the mother.

SIDE EFFECTS

In general, treatment with losartan-hydrochlorothiazide is well tolerated. Side effects have usually been mild and transient in nature and have not required discontinuation of therapy. The most common drug-related side effects were dizziness, asthenia/fatigue and vertigo.
The following adverse reactions have been reported in post-marketing experience:
Hypersensitivity: Anaphylactic reactions, angioedema including swelling of the larynx and glottis causing airway obstruction and/or swelling of the face, lips, pharynx, and/or tongue have been reported rarely in patients treated with losartan; some of these patients previously experienced angioedema with other drugs including ACE inhibitors. Vasculitis, including Henoch-Schoenlein purpura, has been reported rarely with losartan.
Gastrointestinal: Hepatitis (reported rarely in patients treated with losartan), diarrhea.
Respiratory: Cough has been reported with losartan.
Skin: Urticaria, erythroderma has been reported with losartan.
Additional side effects that have been seen with one of the individual components and may be potential side effects with losartan-hydrochlorothiazide are the following:

Losartan
Dose-related orthostatic effects, liver function abnormalities, myalgia, migraine, rash, anaemia, pruritus.

Hydrochlorothiazide
Anorexia, gastric irritation, nausea, vomiting, cramping, diarrhoea, constipation, jaundice (intrahepatic cholestatic jaundice), pancreatitis, sialoadenitis, vertigo, paraesthesiae, headache, xanthopsia, leucopenia, agranulocytosis, thrombocytopenia, aplastic anaemia, haemolytic anaemia, purpura, photosensitivity, fever, necrotising angiitis, respiratory distress (including pneumonitis and pulmonary oedema), anaphylactic reactions, toxic epidermal necrolysis, hyperglycaemia, glycosuria, hyperuricaemia, electrolyte imbalance (including hyponatraemia and hypokalaemia), renal dysfunction, interstitial nephritis, renal failure, muscle spasm, weakness, restlessness, transient blurred vision.

Neoplasms benign, malignant and unspecified (incl cysts and polyps)
Frequency 'not known': Non-melanoma skin cancer (Basal cell carcinoma and Squamous cell carcinoma).
Non-melanoma skin cancer: Based on available data from epidemiological studies, cumulative dose-dependent association between hydrochlorothiazide and nonmelanoma skin cancer has been observed.

SYMPTOMS AND TREATMENT FOR OVERDOSAGE

No specific information is available on the treatment of overdosage with losartan-hydrochlorothiazide. Treatment is symptomatic and supportive. Therapy with losartan-hydrochlorothiazide should be discontinued and the patient observed closely. Suggested measures include induction of emesis if ingestion is recent and correction of dehydration, electrolyte imbalance, hepatic coma, and hypotension by established procedures.

Losartan
The most likely manifestation of overdosage would be hypotension and tachycardia; bradycardia could occur from parasympathetic (vagal) stimulation. If symptomatic hypotension should occur, supportive treatment should be instituted.
Neither losartan nor the active metabolite can be removed by haemodialysis.

Hydrochlorothiazide
The most common signs and symptoms observed are those caused by electrolyte depletion (hypokalaemia, hypochloraemia, hyponatraemia) and dehydration resulting from excessive diuresis. If digitalis has also been administered, hypokalaemia may accentuate cardiac arrhythmias. The degree to which hydrochlorothiazide is removed by haemodialysis has not been established.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

No studies on the effects on the ability to drive and use machines have been performed. However, when driving vehicles or operating machinery it must be borne in mind that dizziness or drowsiness may occasionally occur when taking antihypertensive therapy, in particular during initiation of treatment or when the dose is increased.

STORAGE CONDITION

Store below 30°C. Protect from light.

SHELF-LIFE

The expiry date is indicated on the packaging.

PRODUCT DESCRIPTION

Zylovaa-H Tablet 50/12.5mg:
Yellow, oval (11mm x 6mm), normal convex film-coated tablet with 'Z' marking on one side and plain on reverse. Available as blister strips of 10's in packing of 30 tablets per box.

Zylovaa-H Tablet 100/25mg:
Yellow, oval (15mm x 8mm), normal convex film-coated tablet with 'Z' marking on one side and plain on reverse. Available as blister strips of 10's in packing of 30 tablets per box.

KEEP OUT OF REACH OF CHILDREN/ JAUHI DARI KANAK-KANAK

For further information, please consult your pharmacist or physician.

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Manufacturer and Product Registration Holder:
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